

3	A uniform spherical star has a mass of 6.0×10^{30} kg. The mass of the star may be assumed to be a point mass at the centre of the star. The star may be considered to be isolated in space.		
	(a)	Show that the gravitational field strength at a point 3.0×10^9 m from the centre of the spherical star is 44.5 N kg^{-1}.	
			[1]
	L1	$g = \frac{GM}{r^2} = \frac{(6.67 \times 10^{-11})(6.0 \times 10^{30})}{(3.0 \times 10^9)^2}$ $= 44.467 = 44.5 \text{ Nkg}^{-1}$	M1 A0
	(b)	The radius of the star is 1.0×10^9 m. On the axes of Fig. 3.1, sketch a graph to show the variation with distance x from the centre of the star of the gravitational field strength g of the star for values of x from $x = 1.0 \times 10^9$ m to $x = 4.0 \times 10^9$ m.	

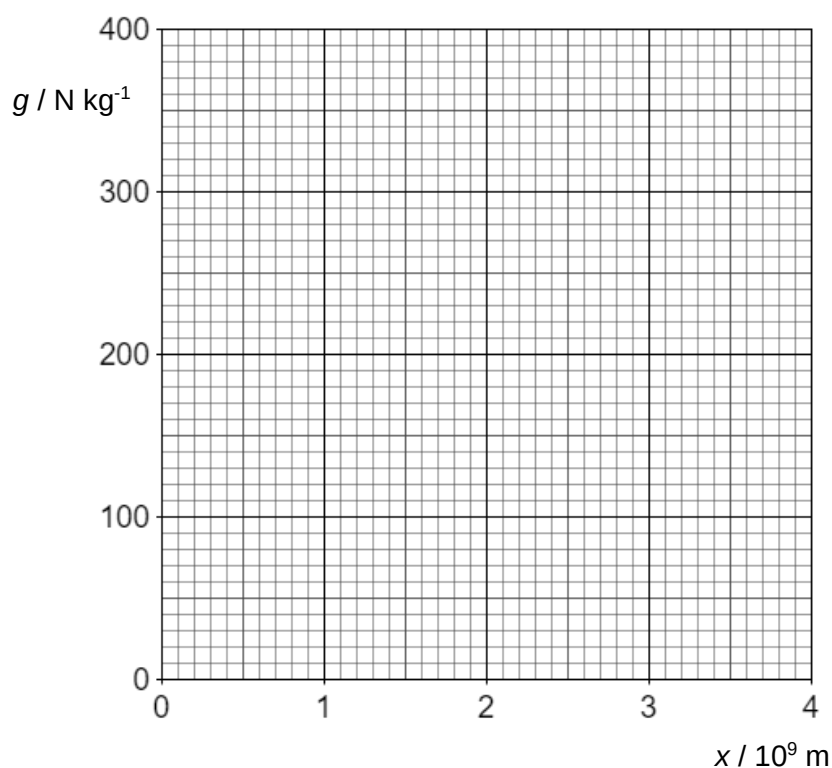


Fig. 3.1

[3]

L2

$$g = \frac{GM}{x^2}$$

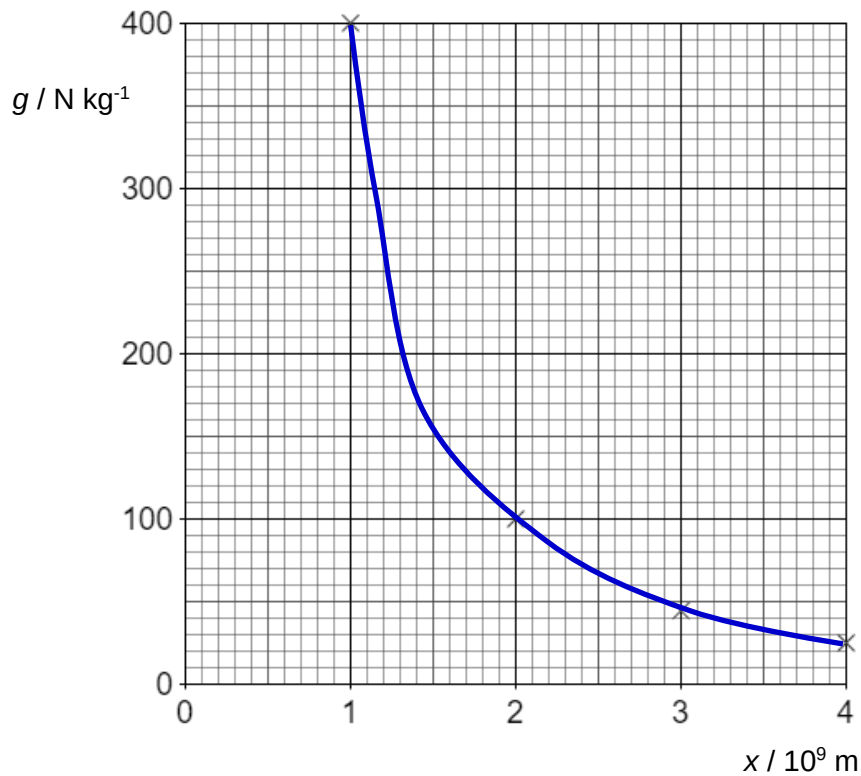
Thus **g** is inversely proportional to r^2 .

When $x = 1.0 \times 10^9$ m,

$$g = \frac{GM}{x^2} = \frac{(6.67 \times 10^{-11})(6.0 \times 10^{30})}{(1.0 \times 10^9)^2} = 400.2 \text{ N kg}^{-1}$$

OR, using (a), $g = (3^2)(44.467) = 400.2 \text{ N kg}^{-1}$

x / m	$g / \text{N kg}^{-1}$
1.0×10^9	400.2
2.0×10^9	$\frac{1}{4} \times 400.2 = \mathbf{100.1}$
3.0×10^9	From (a): 44.5
4.0×10^9	$\frac{1}{4} \times 100.1 = \mathbf{25.0}$



B2

2 marks – At least 4 correctly plotted points in the required range from $x = 1.0 \times 10^9 \text{ m}$ to $x = 4.0 \times 10^9 \text{ m}$, and 4 points relatively well spaced out.

- [1 mark if 1 point wrongly plotted.]
- [0 mark if 2 or more points wrongly plotted or no point plotted.]

1 mark – Smooth hyperbolic curve drawn in the required range from $x = 1.0 \times 10^9 \text{ m}$ to $x = 4.0 \times 10^9 \text{ m}$.

B1

(c) State what the **area under** the graph in Fig. 3.1 represents.

L1 Change in gravitational potential (from $x = 1.0 \times 10^9 \text{ m}$ to $x = 4.0 \times 10^9 \text{ m}$).

OR

Work done per unit mass by an external agent in moving a small mass from $x = 1.0 \times 10^9 \text{ m}$ to $x = 4.0 \times 10^9 \text{ m}$.

[1]

B1

[Total: 5]